Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 6: Physical and Chemical Changes

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	6 — Physical and Chemical Changes
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Traps: 'irreversible therefore chemical' (false), 'dramatic/energetic therefore chemical' (false), 'dissolving/state-change makes a new substance' (false), confusing the white ash mass vs ribbon mass, treating crystallisation as chemical, treating galvanising's sacrificial action as mere physical sealing, and reading 'colour change' as automatic proof of chemistry. Do not write on this paper — mark answers on your answer sheet.

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1. A teacher writes: 'A change in which one or more new substances are formed.' A student replies that this is also true of dissolving sugar, because sugar-water tastes different. The single best reason the student is wrong is that:

(A) sugar-water is colourless, and colourless liquids cannot contain new substances
(B) dissolving always absorbs heat, and only heat-releasing changes can be chemical
(C) evaporating the water gives back the very same sugar, so no substance with new chemical identity was ever formed
(D) the sugar particles become too small to see, which proves they were destroyed

2. Equal masses of magnesium ribbon are burnt completely in air; the white ash is weighed. Compared with the original ribbon, the ash:

(A) weighs less, because burning always destroys part of the metal as smoke
(B) weighs more, because the magnesium has combined with oxygen from the air to form magnesium oxide
(C) weighs exactly the same, because mass cannot change in any reaction
(D) weighs less, because the dazzling light carries away some of the metal's mass

3. Four learners give a 'sure test' for a chemical change. Which test is reliable on its own?

(A) Check whether the change gave out or took in heat

(B) Check whether the substance changed colour

(C) Check whether the change cannot be reversed

(D) Check whether a substance with new properties has formed that cannot be undone by a simple physical method

4. A student argues: 'Melting wax is reversible, so all reversible changes are physical, and burning wax is irreversible, so all irreversible changes are chemical.' Which single example most directly breaks the second half of this rule?

(A) Breaking a ceramic cup — irreversible by simple means, yet no new substance forms, so it is physical

(B) Boiling water into steam — reversible, so it supports the rule

(C) Cooking an egg — irreversible and chemical, so it supports the rule

(D) Freezing water into ice — reversible, so it supports the rule

5. Carbon dioxide is bubbled into clear lime water, turning it milky. If excess CO_2 continues to be passed for a long time, the milkiness can disappear again. The milkiness itself is caused by:

(A) tiny CO_2 bubbles scattering light, a purely physical cloudiness

(B) the lime water freezing into fine ice crystals

(C) a white insoluble substance (calcium carbonate) forming as a precipitate, signalling a chemical change

(D) dust from the air settling on the surface of the liquid

6. An iron nail is dropped into blue copper sulphate solution. After an hour, a reddish-brown coating clings to the nail and the solution looks pale green. The reddish-brown coating is:

(A) rust, because the nail has begun to corrode in the liquid

(B) a film of paint that was already on the nail

(C) copper metal, displaced from the solution by the more reactive iron

(D) iron sulphate that has stuck to the nail's surface

7. Three identical bare iron nails sit in three tubes: Tube A, boiled water sealed under a layer of oil; Tube B, ordinary tap water open to air; Tube C, a tube of dry air kept with a drying agent. After a week, which nail(s) rust?

(A) Only the nail in Tube B, because it alone has both air (oxygen) and water present

(B) All three nails, because iron always rusts in time

(C) Only the nail in Tube C, because dry air is rich in oxygen

(D) The nails in Tube A and Tube C, but not Tube B

8. Why does a scratch through the zinc coat of a galvanised iron sheet still leave the iron protected, whereas a scratch through ordinary paint soon lets rust start underneath?

(A) Zinc heals itself by instantly growing back over the scratch

(B) Zinc is more reactive than iron, so it corrodes in preference and goes on shielding the exposed iron even where the coat is broken

(C) Zinc is harder than paint, so scratches simply cannot reach the iron

(D) The scratch in zinc lets in less air than a scratch in paint

9. Hot saturated copper sulphate solution is left to cool slowly and blue crystals grow. A student labels this 'a chemical change because the colour intensified.' The correct verdict is:

(A) It is a chemical change, because deepening blue means a new compound formed

(B) It is a chemical change, because heat was needed to dissolve the solid first

(C) It is neither physical nor chemical, because only the water moved

(D) It is a physical change (crystallisation); the same copper sulphate simply comes out of solution as pure crystals

10. Which everyday process is physical even though it looks energetic or permanent?

(A) A sparkler showering coloured sparks as it burns

(B) Solid carbon dioxide (dry ice) turning straight into a gas as it 'smokes'

(C) Milk slowly turning into thick curd overnight

(D) An iron railing developing a brown flaky coating outdoors

11. A nail kept in tube X (water boiled to drive off dissolved air, then sealed under oil) does not rust, while a nail in tube Y (plain tap water open to air) does. Together these two tubes are designed to prove that rusting needs:

(A) dissolved air (oxygen), since the only factor tube X removed while keeping water present was the air

(B) water, since tube X clearly had no water at all

(C) heat, since tube X's water had been boiled

(D) salt, since tube Y must have contained dissolved salt

12. Which list is correctly ordered as physical, chemical, chemical?

(A) Burning of a candle wick; melting candle wax; milk souring

(B) Dissolving salt in water; crystallising salt back; baking bread

(C) Souring of milk; melting wax; rusting of a nail

(D) Melting candle wax; the candle wick burning; milk souring

13. Riya says, 'Any change that gives out heat must be chemical.' To show she is wrong using a PHYSICAL change, the best demonstration is:

(A) burning a strip of magnesium and feeling intense heat

(B) stirring anhydrous solid into water and feeling the beaker warm up, yet recovering the same solid on evaporation

(C) letting wood burn in a fire and feeling it radiate heat

(D) watching quicklime react with water to form a new substance with heat

14. A change shows ALL of: a new colour, a gas bubbling off, AND it cannot be reversed by any simple physical step. The safest classification is:

(A) a physical change, because colour changes are always physical

(B) a chemical change, since several independent clues point to a new substance forming

(C) merely a change of state, because a gas appeared

(D) crystallisation, because something new became visible

15. When green leaves carry out photosynthesis, sunlight drives the making of glucose and oxygen from carbon dioxide and water. This is classed as chemical, NOT physical, chiefly because:

(A) the leaf changes to a deeper shade of green during the day

(B) it can be reversed simply by placing the plant in the dark

(C) water merely changes from liquid to vapour inside the leaf

(D) new substances (glucose and oxygen) with different properties are produced from the starting materials

16. An object is found to rust faster than an identical one. Which single difference would speed up rusting the MOST?

(A) keeping it splashed with salty water in open air

(B) keeping it in a sealed jar with silica-gel drying packets

(C) keeping it in air that has had all its moisture removed

(D) coating it thinly with grease and storing it indoors

17. Which pair correctly matches the change to its type?

(A) Curdling of milk — physical; melting of ghee — chemical

(B) Rusting of a chain — physical; cutting of cloth — chemical

(C) Sublimation of camphor — physical; ripening of a mango — chemical

(D) Digestion of food — physical; freezing of water — chemical

18. A goldsmith hammers a lump of gold into a thin sheet and then into a ring. Throughout, the metal stays gold. The shaping is:

(A) a chemical change, because hammering forces the metal to react

(B) a chemical change, because the gold becomes harder

(C) crystallisation, because the metal forms a new ordered shape

(D) a physical change, because only the shape alters while the substance remains gold

19. To get pure copper sulphate crystals from a sample mixed with insoluble sand, the correct sequence is:

(A) heat the sample strongly to burn off the sand, then collect what is left

(B) leave the sample in damp air until crystals rust out of it

(C) dissolve in hot water, filter out the sand, then cool the clear solution slowly so crystals grow

(D) grind the sample finely and pick the blue grains out by hand

20. Magnesium ribbon is often cleaned with sandpaper before it is burnt. The cleaning step is a physical change, but the burning step is chemical. The KEY difference is that during burning:

(A) the ribbon merely melts and then re-solidifies as the same magnesium

(B) the sandpaper marks rub off, exposing fresh shiny metal

(C) the ribbon is only cut into smaller, brighter pieces

(D) magnesium combines with oxygen to form a new white substance (magnesium oxide) with different properties

21. A student lists clues of a chemical change: (i) a precipitate forms, (ii) the object is cut into pieces, (iii) a gas is given off, (iv) heat or light is produced. How many are genuine clues of a chemical change?

(A) four
(B) three
(C) two
(D) one

22. Compared with the shiny magnesium ribbon, the white magnesium oxide ash left after burning:

(A) has exactly the same properties as the original metal

(B) will turn back into shiny ribbon if left to cool overnight

(C) is still pure magnesium, just powdered

(D) has new, different properties and cannot be turned back into the ribbon by simply cooling it

23. Which change would BOTH give off a gas AND be a chemical change?

(A) baking soda reacting with vinegar to fizz(B) boiling water until it turns to steam

(C) opening a fizzy drink so the bubbles escape

(D) solid camphor subliming directly into vapour

24. Two clear solutions are mixed and a solid suddenly appears, settling at the bottom. This solid is called a precipitate, and its appearance tells us that:

(A) the liquids simply got too cold and one froze

(B) a chemical reaction has formed a new insoluble substance

(C) dust from the air happened to settle in the beaker

(D) one liquid evaporated, leaving its crystals behind

25. Galvanising and painting both fight rust, but in DIFFERENT ways. Which statement is correct?

(A) Both protect only by forming a barrier, so a scratch ruins both equally

(B) Paint protects chemically and galvanising protects only as a barrier

(C) Galvanising works by turning the iron itself into zinc

(D) Paint protects only as a barrier, while galvanising's zinc also corrodes in place of the iron, protecting it even at a scratch

26. Burning of petrol in an engine and rusting of an iron gate are sometimes compared. Which statement is correct?

(A) Burning is chemical but rusting is just a physical coating

(B) Both are physical changes because the metal and fuel stay the same substances

(C) Both are chemical changes in which a substance combines with oxygen, but burning is fast while rusting is slow

(D) Rusting is chemical but burning is only a change of state of the fuel

27. A wet shirt dries on a line; later, the same shirt left in a heap with food stains develops a sour smell. The drying and the souring are, respectively:

(A) a physical change (water evaporating) and a chemical change (new smelly substances forming)

(B) both physical changes, because the shirt looks unchanged

(C) both chemical changes, because the shirt's fibres react each time

(D) a chemical change (drying) and a physical change (souring)

28. Why do snack makers seal chips in moisture-free, airtight pouches, sometimes with a tiny 'do not eat' sachet inside?

(A) to keep out oxygen and moisture, slowing the chemical changes that spoil the food

(B) to add extra oxygen so the food stays fresher for longer

(C) to make the food rust, which keeps it crunchy

(D) to physically squash the chips so fewer fit, saving money

29. Which set lists ONLY physical changes?

(A) Burning of wood, melting of wax, dissolving of sugar

(B) Cooking of rice, freezing of water, bending of wire

(C) Rusting of iron, breaking of glass, tearing of cloth

(D) Melting of ice, dissolving of salt, cutting of paper

30. A pure sample of common salt is dissolved in water and the water is then fully evaporated. The white solid left behind is:

(A) a brand-new compound made by mixing salt with water

(B) the same common salt, recovered unchanged — showing the whole process was physical

(C) a fresh batch of crystals chemically different from the salt put in

(D) rust formed while the water sat in the dish

31. An iron nail reacts in copper sulphate solution. After the reaction the liquid is pale green. The green colour is due to:

(A) copper metal that dissolved and coloured the liquid green

(B) iron sulphate, a new substance formed as iron went into solution

(C) rust particles floating in the solution

(D) the original copper sulphate simply fading in the light

32. Which is the odd one out — the ONLY physical change in the list?

(A) Curdling of milk into yoghurt (B) Burning of a candle's wick

(C) Ripening of a green tomato to red

(D) Condensation of water vapour into dew on a cold morning

33. A 'displacement' chemical change is best illustrated by:

(A) candle wax melting on a hot afternoon(B) salt crystals dissolving in warm tea

(C) water vapour condensing on a cold glass

(D) an iron nail pushing copper out of copper sulphate solution

34. Why is freshly made rust especially damaging to an iron railing over the years?

(A) Rust forms a hard, shiny shield that protects the iron from further attack

(B) Rust makes the iron stronger and heavier, so the railing improves

(C) Rust is flaky and crumbles away, exposing fresh iron underneath so the rusting keeps spreading

(D) Rust conducts heat away, cooling the iron and stopping any further change

35. Which everyday cooking step is a chemical change?

(A) Slicing a loaf of bread into pieces

(B) Bread dough baking and browning into a loaf

(C) Grinding wheat grains into flour (D) Chilling cooked soup in the refrigerator

36. Solar salt pans use the Sun to obtain salt from seawater. The Sun's role is to:

(A) chemically change the salt into a brand-new compound

(B) melt the salt so it can be poured off

(C) evaporate the water so that the dissolved salt is left behind as crystals

(D) make the salt react with oxygen in the air

37. A magnet picks up a heap of iron filings; when the magnet is removed, the filings drop and look exactly as before. Magnetising and then de-magnetising the iron is:

(A) a physical change, because the iron forms no new substance and is unchanged afterwards

(B) a chemical change, because the iron gains magnetic 'energy'

(C) rusting, because the iron has been disturbed in air

(D) crystallisation, because the filings line up in neat rows

38. A blacksmith heats an iron bar red-hot, bends it into a hook, and lets it cool. Ignoring any thin surface scale, the bending is:

(A) a chemical change, because the iron was heated until it glowed

(B) rusting, because hot iron always rusts as it cools

(C) a physical change, because the bar is still iron, only its shape is different

(D) crystallisation, because the cooling metal forms crystals

39. At night, leaves respire: they take in oxygen and break down sugar, releasing carbon dioxide and energy. Respiration is a chemical change because it:

(A) breaks down sugar into new substances (carbon dioxide and water) and releases energy

(B) is only a change of state of the sugar inside the leaf

(C) can be reversed simply by watering the plant

(D) produces no new substance, only warmth

40. A student says, 'A change that releases light and great heat, like a burning sparkler, must be chemical.' Is the reasoning sound?

(A) Yes, because any release of light or heat by itself always proves a chemical change

(B) No, because light and heat can never accompany a chemical change

(C) Yes here, because the sparkler forms new substances; but light and heat alone are not always proof, so the real test is whether a new substance forms

(D) No, because burning a sparkler is actually a physical change of state

41. Four jars each hold an identical iron nail: (P) dry air with a drying agent; (Q) boiled water sealed under oil; (R) ordinary tap water open to air; (S) salty water open to air. Which nail rusts the MOST?

(A) Nail P, because dry air is full of oxygen

(B) Nail S, because salty water plus open air gives oxygen, plenty of moisture, and salt that speeds corrosion

(C) Nail Q, because the oil traps heat that drives rusting

(D) Nail R, because tap water rusts iron faster than salt water

42. Which statement about physical and chemical changes is TRUE?

(A) A new substance forms in a chemical change but not in a physical change

(B) Every physical change is reversible and every chemical change is irreversible

(C) Only chemical changes can ever give out heat

(D) Physical changes always produce a precipitate

43. Chrome plating on a steel cycle handle and a sheet of stainless steel both resist rust. The shared reason they resist rusting is that:

(A) both turn the underlying iron into pure gold

(B) both keep a rust-resistant surface (a chromium-containing layer or alloy) between the iron and air and moisture

(C) both make the iron react faster with oxygen, using it up

(D) both work only because they are always kept under water

44. Which observation, taken ON ITS OWN, is the WEAKEST evidence that a chemical change has occurred?

(A) a gas with a sharp new smell was released

(B) a white insoluble precipitate suddenly settled out

(C) the substance simply changed from a solid to a liquid

(D) the mixture glowed and gave out strong heat and light

45. Sublimation of camphor, melting of ice, and bending of a metal spoon are grouped together by a student. What do all three share?

(A) In each, a new substance forms, so all three are chemical changes

(B) In each, the substance keeps its identity, so all three are physical changes

(C) Each gives off a gas, so all three are chemical changes

(D) Each forms a precipitate, so all three are chemical changes

46. The word equation that correctly describes rusting is:

(A) iron + carbon dioxide → rust (B) iron + nitrogen → rust

(C) iron + salt → rust, with no air or water needed

(D) iron + oxygen + water → hydrated iron oxide (rust)

47. Painting an iron tank inside and greasing it outside both prevent rust. The shared mechanism is that paint and grease both:

(A) chemically turn the surface iron into zinc

(B) dry the iron out so completely that it can never react

(C) add a coat of rust that then protects the metal

(D) form a barrier that keeps air and moisture from reaching the iron surface

48. Which change is reversible by a simple physical means?

(A) burning a matchstick to ash (B) baking a cake from raw batter

(C) the souring of milk into curd

(D) boiling water to steam and then condensing it back to water

49. A teacher demonstrates that limewater turns milky with CO_2. A student suggests blowing exhaled breath through a straw into fresh limewater. The expected result and reason are:

(A) the limewater turns milky, because exhaled breath contains carbon dioxide that reacts to form a white solid

(B) the limewater turns blue, because breath adds oxygen

(C) nothing happens, because breath contains only water vapour

(D) the limewater turns clear and fizzes, because breath is mostly nitrogen

50. Two changes are compared: (i) sugar caramelising in a hot pan to a brown, bitter substance, and (ii) sugar dissolving in cold water. The correct classification is:

(A) both are physical, because both involve only sugar

(B) both are chemical, because the sugar changes in each

(C) caramelising is chemical (a new brown substance forms) and dissolving is physical (sugar is recoverable)

(D) caramelising is physical and dissolving is chemical

51. A controlled rust experiment must isolate one factor at a time. To test whether MOISTURE is needed (keeping air available), the correct pair of tubes is:

(A) one nail in open dry air with a drying agent versus one nail in open moist air — both have air, differing only in moisture

(B) one nail in boiled water under oil versus one in dry air — differing in everything

(C) one nail in salty water versus one in plain water — both wet

(D) one nail painted versus one greased — both sealed from air

52. Which statement about crystallisation is CORRECT?

(A) It is a physical change used to obtain pure crystals of a substance from its solution

(B) It is a chemical change because new crystals are a new substance

(C) It works only by burning the solution to dryness

(D) It always releases a gas as the crystals form

53. Cooking a raw egg until it sets is irreversible, yet the change of milk into curd is also irreversible. A student concludes 'both are chemical because both are irreversible.' The conclusion happens to be right, but the REASONING is flawed because:

(A) irreversibility alone does not prove a change is chemical; the real proof is that new substances form, which they do here

(B) irreversibility always guarantees a chemical change, so the reasoning is perfectly fine

(C) both changes are in fact physical, so the conclusion is wrong

(D) neither egg-cooking nor curdling forms a new substance

54. A nail kept in tube A (dry air, drying agent) and a nail in tube B (water that was boiled and sealed under oil) BOTH fail to rust over a week. Together these two tubes show that:

(A) rusting needs only air, since tube A had air yet did not rust

(B) rusting needs BOTH air and moisture, since tube A lacks moisture and tube B lacks air, and neither rusts

(C) rusting needs only moisture, since tube B had water yet did not rust

(D) iron cannot rust under any laboratory conditions

55. A laboratory note lists four observations from one experiment: 'a white precipitate appeared, a gas bubbled off, a sharp new smell arose, and a lot of heat was released.' How many of these are recognised clues of a chemical change?

(A) four

(B) three

(C) two

(D) one

56. Which comparison of dissolving common salt versus burning a candle is CORRECT?

(A) Both are chemical changes because both make the substance disappear

(B) Dissolving salt is chemical and burning a candle is physical

(C) Dissolving salt is physical and reversible by evaporation, while burning a candle is chemical and forms new substances

(D) Both are physical changes that are easily reversed

57. A change is observed to give out heat, change colour, and form a gas, yet a student insists 'it might still be physical.' What single follow-up test would best settle whether it is chemical?

(A) Measure exactly how much heat was given out and compare it to a table

(B) Try to recover the original substances by a simple physical method; if you cannot, a new substance has formed and it is chemical

(C) Check only whether the colour change was bright enough to see

(D) Smell the gas and decide based on whether it is pleasant or unpleasant

58. A thin iron sheet is dipped into molten zinc to galvanise it. The dipping step itself, before any later corrosion, is best described as:

(A) a chemical change that turns the iron into a brand-new metal

(B) rusting, because hot metal always corrodes during dipping

(C) a physical coating process — a layer of zinc is laid over the iron without changing the iron's identity

(D) crystallisation of the iron out of the molten zinc

59. Which single observation, by itself, would MOST strongly suggest that a chemical change has happened rather than a physical one?

(A) A solid block is sawn into several smaller blocks

(B) A shiny metal is bent into a curved shape

(C) A gas with a completely new smell bubbles out while a colour change appears and the starting material cannot be recovered

(D) A coloured liquid spreads out and becomes paler when more water is added

60. A student claims: 'Because melting ice and freezing water are reversible, every change of state is reversible and therefore physical.' Which case best supports that changes of state are physical?

(A) Wax vapour at a candle flame burning to form carbon dioxide and water

(B) Wood heated until it chars into black charcoal and smoke

(C) Limestone heated strongly until it gives off a gas and leaves a new solid

(D) Steam condensing back into liquid water and then re-boiling into steam, the same water throughout

Solutions — Science (Class 7), Chapter 6: Physical and Chemical Changes — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	B	31	B	41	B	51	A
2	B	12	D	22	D	32	D	42	A	52	A
3	D	13	B	23	A	33	D	43	B	53	A
4	A	14	B	24	B	34	C	44	C	54	B
5	C	15	D	25	D	35	B	45	B	55	A
6	C	16	A	26	C	36	C	46	D	56	C
7	A	17	C	27	A	37	A	47	D	57	B
8	B	18	D	28	A	38	C	48	D	58	C
9	D	19	C	29	D	39	A	49	A	59	C
10	B	20	D	30	B	40	C	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Dissolving is physical: the sugar is still chemically sugar and can be fully recovered by evaporating the water, so no new substance was made — that is the decisive test, not taste. A change is chemical only when a substance of new chemical identity appears. Recovery by a physical means (evaporation) is the proof here. Colourlessness is irrelevant — many solutions of new compounds are colourless. Heat absorption does not decide the issue: some physical changes absorb or release heat. 'Too small to see' is wrong; the particles are merely dispersed, not destroyed.

2. (B) Burning magnesium combines it with oxygen to make magnesium oxide (MgO), so the ash contains the original magnesium PLUS the added oxygen — it is heavier than the ribbon. This is a chemical change (a new substance with new properties). 'Burning destroys metal as smoke' is the common misconception; here the product is a solid that gained oxygen. 'Same mass' confuses the conservation idea (total mass including the air's oxygen is conserved, but the solid alone gains mass). Light does not remove mass.

3. (D) The only dependable signature of a chemical change is the formation of a new substance with new properties, not recoverable by a physical step. Heat exchange happens in many physical changes too (dissolving washing soda warms; melting ice cools). Colour change can be physical (heating a wire glows; some indicators are merely diluted). Irreversibility alone is false — breaking glass is irreversible yet physical. So only the new-substance criterion holds up alone.

4. (A) The rule 'irreversible ⇒ chemical' is false, and the cleanest counter-example is a smashed cup: you cannot reassemble it, yet the pieces are still the same material — no new substance, so it is physical. Boiling and freezing are reversible examples that do not test the irreversible-half claim. Cooking an egg is genuinely irreversible AND chemical, so it agrees with the false rule rather than refuting it. Only the broken cup shows irreversible-but-physical.

5. (C) CO_2 reacts with lime water to form insoluble calcium carbonate, whose fine white particles make the liquid look milky — a precipitate is a classic chemical-change clue. The new solid has different properties from the starting materials. Scattering by gas bubbles is not the cause; the cloudiness is a solid product. No freezing occurs at room temperature. Dust is irrelevant — the milkiness appears the instant CO_2 reacts, in a sealed clean tube too.

6. (C) Iron is more reactive than copper, so it pushes copper out of copper sulphate (a displacement reaction): the freed copper deposits on the nail as a reddish-brown layer, while iron goes into solution forming pale-green iron sulphate. The coating is copper, not rust — rust needs air and water and is a different oxide. There is no paint. Iron sulphate is the green substance dissolved in the liquid, not the solid coat on the nail.

7. (A) Rusting needs iron, oxygen (from air) AND moisture together. Tube A has water but the boiling removed dissolved air and the oil seals air out — no oxygen, so no rust. Tube C has air but no moisture — no rust. Only Tube B supplies both, so only B rusts. 'Iron always rusts' ignores that removing either condition stops it. Dry air alone (C) cannot rust iron because moisture is missing.

8. (B) Galvanising works partly as a barrier but crucially because zinc is more reactive than iron: at a scratch the zinc is attacked first and keeps protecting the bare iron (sacrificial action). Paint is only a barrier — once breached, the exposed iron rusts. Zinc does not magically regrow. Hardness is not why it protects; even a deep scratch is still guarded by the surrounding zinc's chemistry. Air entry is similar for both scratches, so that is not the reason.

9. (D) Crystallisation separates the dissolved copper sulphate as crystals of the very same substance — no new substance forms, so it is physical, used to obtain pure crystals from a solution. The colour deepens only because solid crystals concentrate the blue; no new compound appears. Heating to dissolve does not make it chemical — that step is also physical (dissolving). It is definitely a change (a state/phase change of the solute), so 'neither' is wrong.

10. (B) Dry ice subliming is a change of state — solid CO_2 to gaseous CO_2, same substance — so it is physical despite the dramatic 'smoke'. A sparkler burns metal salts into new substances (chemical). Curdling forms new substances from milk (chemical). The brown coat on iron is rust, a new oxide (chemical). Only sublimation keeps the same substance throughout.

11. (A) Both tubes contain water, so water alone cannot explain the difference. Tube X differs only by having its dissolved air removed (boiling) and sealed off (oil), and it does NOT rust — so the missing oxygen is the cause. Tube X did have water (boiled water is still water), so 'no water' is false. The boiling was to expel air, not to test heat. Salt is not part of this controlled comparison; both used the same water otherwise.

12. (D) Melting wax keeps the same substance (physical); a burning wick makes new substances such as carbon dioxide and water (chemical); souring milk forms new acidic substances (chemical) — so physical, chemical, chemical. The option starting with a burning wick puts chemical first. 'Dissolving, crystallising, baking' is physical, physical, chemical. 'Souring, melting, rusting' is chemical, physical, chemical. Only the first option matches the required order.

13. (B) Some dissolving processes release heat yet form no new substance — the solid can be recovered by evaporation, proving the warm change was physical. That defeats 'heat ⇒ chemical'. Burning magnesium and burning wood are chemical (new substances), so they support Riya rather than refute her. Quicklime plus water forms a brand-new substance (slaked lime) with heat — that is chemical, so it cannot serve as the physical counter-example.

14. (B) A new colour together with gas evolution and irreversibility are mutually reinforcing clues of a new substance — strong evidence for a chemical change. Colour change is NOT always physical (here it joins gas and irreversibility). A gas can also come off in chemical reactions, so its appearance does not make this a mere change of state. Crystallisation produces solid crystals of the same substance, with no gas or new colour, so it does not fit.

15. (D) Photosynthesis builds glucose and releases oxygen — substances with properties unlike CO_2 and water — so a new substance forms, the hallmark of a chemical change. A shade change is incidental, not the defining reason. It is not reversed by darkness (respiration is a separate reaction, not an 'undo'). It is far more than water turning to vapour; chemical bonds are rearranged to make glucose.

16. (A) Salt water in open air supplies oxygen and abundant moisture and the dissolved salt makes the water conduct better, all of which speed up rusting most. Silica-gel removes moisture, so rusting is slowed, not sped up. Dry air lacks the needed moisture, again slowing rust. Grease seals out air and moisture, the slowest case. So salty water in open air is the fastest-rust condition.

17. (C) Camphor subliming is a state change of the same substance (physical); a mango ripening develops new substances such as sugars and aroma compounds (chemical) — both correct. Curdling is chemical and melting ghee is physical, so that pair is swapped. Rusting is chemical and cutting cloth is physical, also swapped. Digestion is chemical and freezing is physical, again swapped. Only the camphor/mango pair is right.

18. (D) Beating gold changes only its shape; it is still gold, so the change is physical. Hammering does not make gold react with anything — no new substance forms. Any hardness change is a physical property shift, still not a new substance. Crystallisation specifically means crystals coming out of a solution, which has nothing to do with shaping solid metal by hammering.

19. (C) Crystallisation purifies by dissolving the soluble copper sulphate in hot water, filtering away the insoluble sand, and cooling slowly so pure crystals separate. Burning does not remove sand and would damage the copper sulphate. Damp air causes rusting of iron, irrelevant to copper sulphate and producing no crystals. Hand-picking cannot separate finely mixed material reliably or give pure crystals; only crystallisation does.

20. (D) Burning bonds magnesium with oxygen to give magnesium oxide — a new white powder that no longer behaves like the shiny metal, so it is chemical. Cleaning, by contrast, only removed a surface layer (physical). The ribbon does not simply melt back into the same magnesium; it is chemically converted. No mere cutting is involved; the substance's identity itself changes.

21. (B) Precipitate formation, gas evolution, and the giving out of heat or light are recognised clues of a chemical change — that is three. Cutting an object into pieces only changes its size and shape (physical), so it is NOT a clue of a chemical change. Hence three, not four. It is more than two and far more than one, because three of the four items are valid chemical-change indicators.

22. (D) The ash is magnesium oxide, a new substance whose properties (white, powdery, no metallic shine) differ from the metal, and cooling will not reverse the reaction — proof of a chemical change. It does not share the metal's properties. It will not spontaneously revert to ribbon. It is not 'just powdered magnesium'; oxygen has chemically combined with the metal to make a different compound.

23. (A) Baking soda and vinegar react to make a new gas (carbon dioxide) plus other new substances — gas evolution from a chemical reaction. Boiling water releases steam, but that is the same water as a gas (physical). A fizzy drink fizzes because dissolved CO_2 escapes — no new substance forms, so physical. Camphor subliming is a physical state change of the same substance. Only the soda–vinegar reaction is chemical and gas-producing.

24. (B) A precipitate is a new insoluble solid produced when two solutions react, so its sudden appearance is a clear chemical-change clue. It is not freezing — the solid forms at room temperature from a reaction, not from cooling. It is not stray dust; the solid forms throughout the mixture instantly on mixing. It is not left-over crystals from evaporation, because no liquid was boiled away; the solid grows out of the reaction itself.

25. (D) Paint is purely a barrier — break it and the bare iron rusts. Galvanising adds a barrier of zinc AND, because zinc is more reactive, the zinc corrodes preferentially and shields the iron even at a scratch. So a scratch does not ruin both equally; galvanising still protects. It is paint that is the simple barrier, not the reverse. Galvanising does not transmute iron into zinc; it merely coats it.

26. (C) Both rusting and burning are chemical changes that involve combining with oxygen — burning is rapid (oxidation with heat and light), rusting is slow oxidation. Rusting forms a new oxide, not a mere coating of the same iron. Neither is physical; in both, new substances appear. Burning is not a change of state — petrol vapour reacts to form new gases, so it is chemical, not just evaporation.

27. (A) Drying is water leaving the cloth as vapour — same water, so physical. The sour smell comes from new substances produced as the stains spoil — that is chemical. They are not both physical: souring makes new compounds. They are not both chemical: evaporation forms no new substance. The labels are not reversed; drying is the physical one and souring the chemical one.

28. (A) Spoiling of food is largely a chemical change driven by oxygen and moisture; sealing them out (the sachet absorbs moisture/oxygen) slows that chemistry and keeps the food fresh. Adding oxygen would speed spoiling, not slow it. Food does not rust — rusting is specific to iron — and rust would not help. The pouch is not meant to crush the chips; airtight, moisture-free packing is about chemistry, not portion size.

29. (D) Melting ice, dissolving salt, and cutting paper each keep the same substance — all physical. The wood-burning set contains burning (chemical), so it fails. The rice set contains cooking (chemical). The rusting set contains rusting (chemical). Only the first set is entirely physical, with no new substance formed in any of the three.

30. (B) Dissolving and then evaporating simply separates the salt back out unchanged — recovery of the original substance proves the change was physical. No new compound forms with water; the salt's identity is preserved. The recovered crystals are chemically the same salt, not a different compound. No rust forms — rusting needs iron, which is not present here.

31. (B) As iron displaces copper, iron enters the solution as iron sulphate, which is pale green — a new substance, confirming a chemical change. Copper does not dissolve to colour the liquid; it deposits as solid copper on the nail. Rust is not formed here; this is a displacement, not air-and-water corrosion. The blue did not merely 'fade in light'; it was chemically replaced by green iron sulphate.

32. (D) Dew forming is water vapour turning to liquid water — same substance, a change of state, so physical. Curdling makes new substances from milk (chemical). A burning wick produces new gases (chemical). A ripening tomato develops new pigments and sugars (chemical). Only condensation keeps the same substance throughout, making it the odd one out.

33. (D) Displacement means a more reactive element pushing out a less reactive one: iron displaces copper from copper sulphate, forming new substances (copper metal and iron sulphate) — chemical. Melting wax is a physical state change. Dissolving salt is a physical mixing, recoverable by evaporation. Condensation is a physical state change. Only the iron-and-copper-sulphate case is a displacement reaction.

34. (C) Rust flakes off rather than clinging tightly, so it does not protect; instead it keeps baring fresh iron to air and moisture, letting corrosion eat deeper over time. It is not a hard protective shield — that is the opposite of what rust does. It weakens, not strengthens, the metal. Cooling has nothing to do with it; the chemistry of oxygen and moisture continues regardless.

35. (B) Baking turns dough into bread with new substances (the browning and crust are new compounds, gas raised it) — a chemical change. Slicing bread only changes its shape (physical). Grinding wheat into flour breaks grains into smaller pieces, the same substance (physical). Chilling soup just lowers its temperature without forming new substances (physical). Only baking is chemical.

36. (C) Sunlight evaporates the seawater, and the salt that was dissolved is left as crystals — a physical separation (a form of crystallisation/evaporation), recovering the same salt. The salt is not chemically changed; it is the same common salt. It is not melted; salt's melting point is far above what the Sun provides. It does not react with oxygen; evaporation involves only the water leaving.

37. (A) Becoming magnetised and losing it again leaves the iron chemically the same — no new substance, so it is physical. Gaining or losing magnetism is not a new substance, so it is not chemical. No rust forms from simply being near a magnet in air for moments. Crystallisation involves crystals from a solution, not filings lining up along magnetic field lines.

38. (C) Bending hot iron alters its shape but not its substance — it remains iron, so the shaping is physical. Glowing from heat is just hot metal radiating light, not a new substance. Rusting needs moisture and air over time; quick shaping and cooling is not rusting. Crystallisation here would refer to crystals from a solution, not the routine cooling of a solid metal bar being shaped.

39. (A) Respiration breaks sugar down into new substances — carbon dioxide and water — while releasing energy, so new substances form and it is chemical. It is not a mere change of state; the sugar molecule is chemically broken apart. Watering does not reverse it. It does make new substances, so 'no new substance, only warmth' is false.

40. (C) The sparkler is chemical because new substances form when its contents burn; light and heat are supporting clues, not the defining test — some physical processes (a glowing filament, dissolving washing soda) also give light or heat. So the correct reasoning rests on the new-substance test. Light/heat alone is not always proof. They certainly CAN accompany chemical changes, so 'never' is wrong. Burning a sparkler is chemical, not a change of state.

41. (B) Salt water open to air (S) supplies both oxygen and abundant moisture, and dissolved salt accelerates rusting, so S rusts the most. P lacks moisture (drying agent), so it barely rusts. Q has water but no air (boiled and oil-sealed), so it does not rust. R rusts, but plain tap water is slower than salty water, so it is not the most. Hence S.

42. (A) The defining truth is that a chemical change makes a new substance while a physical change does not. 'Every physical change reversible / chemical irreversible' is false — breaking glass is physical yet irreversible. 'Only chemical changes give out heat' is false — dissolving some solids releases heat physically. Physical changes do not 'always produce a precipitate'; a precipitate is in fact a chemical-change clue, and most physical changes make none.

43. (B) Chrome plating coats the iron with rust-resistant chromium, and stainless steel is an iron alloy containing chromium that resists corrosion — both keep a protective chromium-bearing surface against air and moisture. Neither turns iron to gold. They do not speed iron's reaction with oxygen — the opposite, they slow corrosion. They protect in ordinary air, not only underwater (water would in fact promote rusting of unprotected iron).

44. (C) A solid melting to a liquid is just a change of state — the weakest 'evidence' because it is typically physical, with no new substance. A new-smelling gas suggests a new substance (stronger clue). A precipitate is a recognised chemical clue. A glow with heat and light points to combustion-type chemistry. So melting is the weakest indicator of chemistry of the four.

45. (B) Subliming camphor (solid to gas), melting ice (solid to liquid), and bending a spoon (shape change) all leave the substance chemically unchanged — they are all physical. No new substance forms in any, so 'all chemical' is wrong. Only camphor 'gives off a gas' and even that is its own vapour, not a chemical gas; ice and the spoon give off no gas. None forms a precipitate. The shared feature is unchanged identity.

46. (D) Rusting needs iron together with oxygen (from air) and water (moisture), giving hydrated iron oxide — the brown rust. Carbon dioxide is not the reactant; that is the lime-water test, not rusting. Nitrogen, though plentiful in air, does not rust iron. Salt can speed rusting but is not itself a reactant and cannot replace the essential air and water — so 'iron + salt with no air or water' is wrong.

47. (D) Paint and grease work as physical barriers, sealing the iron from air and moisture so rusting (which needs both) cannot start. Neither converts iron into zinc — that is galvanising, a different method. They do not permanently 'dry out' the metal; they simply block contact, and if the coat breaks, rust can begin. They certainly do not add protective rust; rust is the very thing they prevent.

48. (D) Boiling and condensing water just moves it between liquid and gas — the same water, reversible by cooling, a physical change. Burning a matchstick makes new substances (ash, gases) that cannot be turned back into a match — chemical and irreversible. Baking forms new substances in the cake — chemical, irreversible. Souring makes new acidic substances in curd — chemical, irreversible. Only the water case is reversible physically.

49. (A) Exhaled breath is rich in carbon dioxide, which reacts with limewater to form an insoluble white solid, turning it milky — the same chemical test, demonstrating a new substance. It does not turn blue; oxygen does not produce that reaction. Breath is not only water vapour — its CO_2 is what reacts. The nitrogen in breath does not react with limewater, so 'clear and fizzing' is wrong; the milkiness is the key result.

50. (C) Caramelising heats sugar until it forms new brown, bitter substances — chemical and not reversible. Dissolving merely disperses sugar in water, and evaporating the water gives the sugar back — physical. They are not both physical: caramelising makes a new substance. They are not both chemical: dissolving makes none. The labels are not reversed; caramelising is the chemical one.

51. (A) To test moisture alone, both tubes must have air and differ only in moisture: a nail in dry air (drying agent) versus one in moist air does exactly that, isolating moisture as the variable. Boiled-water-under-oil versus dry air changes both air and water, so it is not a clean single-factor test. Salty versus plain water both have moisture, so it tests salt, not moisture. Painted versus greased both block air, testing neither factor cleanly.

52. (A) Crystallisation separates a dissolved solid as pure crystals of the same substance — no new substance, so it is physical, and it is used to purify (e.g., copper sulphate, sea salt). The crystals are the SAME substance, so it is not chemical. It does not require burning;

usually the solution is cooled or gently evaporated. No gas is produced as crystals settle out of solution.

53. (A) Cooking an egg and curdling milk are chemical because each forms new substances — that is the valid reason, not the mere fact of irreversibility. Irreversibility does NOT guarantee chemistry (breaking glass is irreversible yet physical), so the student's reasoning is flawed even though the verdict is correct. Both changes are chemical, not physical. Both DO form new substances, so the last option is false.

54. (B) Tube A has air but no moisture and does not rust; tube B has moisture but no air and does not rust — so removing EITHER factor stops rusting, proving both air and moisture are needed. 'Only air' is wrong because tube A had air yet stayed shiny. 'Only moisture' is wrong because tube B had water yet stayed shiny. Iron certainly does rust when both factors are present (e.g., open tap water), so the last option is false.

55. (A) Precipitate formation, gas evolution, a new smell (a new substance), and the release of heat are ALL recognised clues of a chemical change — that is four. None of these four is a mere physical clue like a change of size or shape, so the count is not three, two, or one. All four point toward new substances forming.

56. (C) Dissolving salt is physical — the salt is recoverable by evaporating the water. Burning a candle is chemical — the wax vapour forms new substances such as carbon dioxide and water. 'Both chemical because they disappear' is wrong: disappearing into solution is physical. The labels are not swapped; dissolving is the physical one. They are not both physical, because burning cannot be reversed and makes new substances.

57. (B) The decisive test is whether the original substances can be got back by a physical means: if they cannot, a new substance has formed and the change is chemical. Measuring heat alone does not decide it, since physical changes can exchange heat. Brightness of a colour change is not a reliable criterion. Whether a smell is pleasant or not says nothing about whether a new substance formed; only the recovery test settles it.

58. (C) Galvanising lays a zinc coat over the iron; the iron stays iron beneath, so the coating step is a physical process. It does not transmute the iron into a new metal — no new substance is formed in the iron itself. It is not rusting; dipping in molten zinc forms a protective layer, not rust. Crystallisation refers to crystals from a solution, not coating a sheet with molten metal, so that label is wrong.

59. (C) A new-smelling gas plus a colour change plus failure to recover the starting material together strongly indicate a new substance — chemical. Sawing a block only changes its size and shape (physical). Bending metal changes only shape (physical). Diluting a coloured liquid with water makes it paler but forms no new substance, and the original can be recovered by evaporation (physical). So only the first observation points to chemistry.

60. (D) Water cycling between steam and liquid is a pure change of state — the same water all along, reversible, physical — supporting the claim. Wax vapour burning makes new substances (CO_2 and water), a chemical change, not a simple change of state. Wood charring forms new substances (charcoal, smoke gases), chemical. Heating limestone produces a new gas and a new solid, chemical. Only the steam-water case is a reversible state change.